Math 110 Quiz Review

Notes Ch. 14–18, HW8–10

§1 Review

§1.1 Tools from earlier chapters

V is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Every proof below uses only these facts. Each has its reason, except the fundamental theorem of algebra (it needs analysis).

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.
Why: Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V. Then $Tv_1, \dots, Tv_r$ spans $\operatorname{range} T$ (the Tu_i are 0). It is independent: if $\sum c_j T v_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the u_i, so all $c_j = 0$.

(T2) **For operators, injective $\iff$ surjective $\iff$ invertible.**
Why: $\operatorname{null} T = 0 \iff \dim \operatorname{range} T = n \iff \operatorname{range} T = V$, by (T1) and (T3).

(T3) **Dimension shortcuts.**

- An independent list of n vectors is a basis.
- A subspace of dimension n is all of V.
- If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.

Why: An independent list extends to a basis, and all bases have length n. For the last: a basis of U followed by a basis of W is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length n.

(T4) **Products of operators.** If S, T are injective, so is ST. So if a product sends some $v \neq 0$ to 0, some factor is not injective.
Why: $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T5) **Polynomials in T commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are T-invariant.
Why: Both equal $(pq)(T)$. If $p(T)v = 0$ then $p(T)Tv = Tp(T)v = 0$; and $Tp(T)u = p(T)(Tu)$.

(T6) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z - \lambda_1) \cdots (z - \lambda_m)$ (cited). Also $p(\lambda) = 0 \iff (z - \lambda) \mid p$.
Why: Divide p by $z - \lambda$: the remainder is the constant $p(\lambda)$.

(T7) **Eigenvalues.**

- λ is an eigenvalue $\iff$ $T - \lambda I$ is not injective.
- If $Tv = \lambda v$, then $p(T)v = p(\lambda)v$.
- Eigenvectors for distinct eigenvalues are independent (Prop. 54; also a special case of Prop. 72).
- Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

Why: Existence: take $v \neq 0$. The $n + 1$ vectors $v, Tv, \dots, T^n v$ are dependent, so $p(T)v = 0$ for some $p \neq 0$, and p is not constant since $v \neq 0$. Factor p by (T6); by (T4) some $T - \lambda_j I$ is not injective.

(T8) **Upper-triangular matrices.** $\mathcal{M}(T)$ is upper-triangular in $v_1, \dots, v_n \iff$ every $\operatorname{span}(v_1, \dots, v_j)$ is invariant. Its eigenvalues are its diagonal entries. Over $\mathbb{C}$ every operator has one.
Why: Column j holds the coefficients of Tv_j, so "upper-triangular" means $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$. If $a_{jj} = \lambda$, $T - \lambda I$ maps $\operatorname{span}(v_1, \dots, v_j)$ into $\operatorname{span}(v_1, \dots, v_{j-1})$, so it is not injective (T1). If no $a_{jj} = \lambda$, the columns of $\mathcal{M}(T - \lambda I)$ are independent (in a relation, the right-most column used is the only one with a nonzero entry in its diagonal row). Existence: a Jordan form (Prop. 76) is upper-triangular; Props. 69–76 do not use (T8).

(T9) **Minimal polynomial.** $q(T) = 0 \iff m_T \mid q$. Its roots are exactly the eigenvalues. T is diagonalizable $\iff m_T = (x-\lambda_1)\cdots(x-\lambda_m)$ with distinct $\lambda_i \in \mathbb{F}$.
Why: Divide $q = s\,m_T + r$; if $q(T) = 0$ then $r(T) = 0$, so $r = 0$ by minimality. If $Tv = \lambda v$, $v \neq 0$: $0 = m_T(T)v = m_T(\lambda)v$. If $m_T = (x-\lambda)s$: $s(T) \neq 0$, and for w with $s(T)w \neq 0$, $(T-\lambda I)s(T)w = 0$. The diagonalizability test is notes Prop. 68 (Midterm 2).

(T10) **Block diagonal matrices.** If $V = U \oplus W$ with both T-invariant, then in a basis of U followed by one of W, $\mathcal{M}(T) = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$, and $\operatorname{null} T = \operatorname{null} T|_U \oplus \operatorname{null} T|_W$.
Why: $Tu \in U$ uses only the U-basis, so its column is 0 in the W-rows. If $T(u+w) = 0$ then $Tu = -Tw \in U \cap W = \{0\}$.

(T11) **Change of basis.** If column j of S holds the B-coordinates of the j-th vector of B', then $\mathcal{M}_{B'}(T) = S^{-1}\mathcal{M}_B(T)S$. So similar matrices are exactly the matrices of one operator in two bases.
Why: S converts B'-coordinates to B-coordinates, and every invertible S arises this way (its columns form a basis).

§1.2 Generalized eigenvectors (Ch. 14)

The question. We want the simplest matrix for every operator. With an eigenbasis it is diagonal, but often there are too few eigenvectors.

- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has eigenvectors only on $\operatorname{span}(e_1)$.
- But $Ne_2 = e_1$ and $N^2 e_2 = 0$: e_2 is killed after two steps instead of one.
- Idea of the chapter: allow "killed after several steps". Over $\mathbb{C}$ that gives enough vectors for a basis.

Definition (Def. 37 / LADR 8.8, 8.19)**.** $v \neq 0$ is a **generalized eigenvector** for λ if $(T-\lambda I)^k v = 0$ for some $k \geq 1$. The **generalized eigenspace** is $G(\lambda, T) = \bigcup_{k \geq 0} \operatorname{null}(T-\lambda I)^k$.

- $E(\lambda, T) \subseteq G(\lambda, T)$ (take $k = 1$). In the example, $E(0, N) = \operatorname{span}(e_1)$ but $G(0, N) = \mathbb{F}^2$.
- A union over all k is awkward. Props. 69–70 show it stops at a fixed power.

Proposition (Prop. 69 / LADR 8.1, 8.2: the null-space chain)
$\{0\} = \operatorname{null} T^0 \subseteq \operatorname{null} T \subseteq \operatorname{null} T^2 \subseteq \cdots$. If $\operatorname{null} T^m = \operatorname{null} T^{m+1}$, then $\operatorname{null} T^m = \operatorname{null} T^{m+k}$ for all $k \geq 0$.

How to find it

- Growing: once a vector is killed, applying T again still gives 0.
- Not growing again: it suffices to show one step, $\operatorname{null} T^{m+k+1} \subseteq \operatorname{null} T^{m+k}$.
- Key trick: we only know about T^m and T^{m+1}, so move powers of T onto the vector: $T^{m+k+1}v = T^{m+1}(T^k v)$.

Proof. 1. If $T^k v = 0$, then $T^{k+1}v = T(0) = 0$. So $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$.
2. Suppose $\operatorname{null} T^m = \operatorname{null} T^{m+1}$ and $T^{m+k+1}v = 0$. Then $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$.
3. So $T^{m+k}v = T^m(T^k v) = 0$. Hence $\operatorname{null} T^{m+k+1} \subseteq \operatorname{null} T^{m+k}$, and with step 1 they are equal. Induct on k.

□

Proposition (Prop. 70 / LADR 8.3, 8.20)
$\operatorname{null} T^n = \operatorname{null} T^{n+1}$. Hence $G(\lambda, T) = \operatorname{null}(T-\lambda I)^n$, a T-invariant subspace.

How to find it

- Each strict growth raises the dimension by at least 1, and dimensions are at most n.
- So the chain grows at most n times; by Prop. 69 it then stays still.

Proof. 1. Suppose $\operatorname{null} T^n \neq \operatorname{null} T^{n+1}$. By Prop. 69 no earlier pair is equal either.

2. Then $0 = \dim \operatorname{null} T^0 < \dim \operatorname{null} T^1 < \cdots < \dim \operatorname{null} T^{n+1} \leq n$: that is $n+2$ distinct integers in $\{0, \ldots, n\}$. Impossible.
3. So $\bigcup_k \operatorname{null}(T - \lambda I)^k = \operatorname{null}(T - \lambda I)^n$. It is invariant by (T5).

□

Proposition (LADR 8.4, 8.5: splitting off the nilpotent part)
$V = \operatorname{null} T^n \oplus \operatorname{range} T^n$, and T is injective on $\operatorname{range} T^n$.

How to find it

- Goal: separate what T eventually kills ($\operatorname{null} T^n$) from what survives ($\operatorname{range} T^n$).
- Rank–nullity gives the right total dimension, so by (T3) only the intersection needs checking.
- For the intersection, use Prop. 70: anything killed by a high power is killed by T^n.

Proof. 1. Let $v = T^n u$ with $T^n v = 0$. Then $T^{2n} u = 0$, so $u \in \operatorname{null} T^{2n} = \operatorname{null} T^n$, so $v = 0$.
2. $\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1), so $V = \operatorname{null} T^n \oplus \operatorname{range} T^n$ (T3).
3. If $w \in \operatorname{range} T^n$ and $Tw = 0$, then $w \in \operatorname{null} T \subseteq \operatorname{null} T^n$, so $w = 0$.

□

Proposition (Prop. 71)
$G(\lambda, T) \neq \{0\} \iff \lambda$ is an eigenvalue of T.

How to find it

- Generalized eigenvectors cannot exist without ordinary ones: keep applying $T - \lambda I$ to a nonzero $v \in G(\lambda, T)$, stopping just before 0. The last nonzero vector is an eigenvector.

Proof. If λ is an eigenvalue, $\{0\} \neq E \subseteq G$. If not, $\operatorname{null}(T - \lambda I) = \{0\} = \operatorname{null}(T - \lambda I)^0$, so every $\operatorname{null}(T - \lambda I)^k = \{0\}$ (Prop. 69). □

Proposition (Prop. 72 / LADR 8.12)
Generalized eigenvectors $v_1, \ldots, v_m$ for distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ are independent.

How to find it

- Given $\sum a_i v_i = 0$, to get $a_k = 0$ find an operator that kills every v_i $(i \neq k)$ but not v_k.
- $(T - \lambda_i I)^n$ kills v_i, so take the product over $i \neq k$.
- To see its effect on v_k, first push v_k down to an actual eigenvector w. On w, $(T - \lambda_i I)^n$ is the nonzero number $(\lambda_k - \lambda_i)^n$.

Proof. 1. Fix k. Let N be the largest exponent with $w = (T - \lambda_k I)^N v_k \neq 0$. Then $Tw = \lambda_k w$, so $(T - \lambda_i I)^n w = (\lambda_k - \lambda_i)^n w$.
2. Apply $S = \prod_{i \neq k} (T - \lambda_i I)^n \cdot (T - \lambda_k I)^N$ to $\sum a_i v_i = 0$. The factors commute (T5), so $S v_i = 0$ for $i \neq k$.
3. What is left: $0 = a_k \prod_{i \neq k} (\lambda_k - \lambda_i)^n \, w$. The product and w are nonzero, so $a_k = 0$.

□

Theorem (Prop. 73 / LADR 8.22: generalized eigenspace decomposition)
Over $\mathbb{C}$, if $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T, then $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$.

How to find it

- Two things: the G's span V, and the sum is direct. Direct is Prop. 72.
- Spanning: peel off one G at a time. The splitting result gives $V = G(\lambda_1, T) \oplus \operatorname{range}(T - \lambda_1 I)^n$; the second piece is invariant and smaller, so induct.
- $\mathbb{C}$ is needed to have an eigenvalue λ_1 to start with (T7).

Proof. 1. Induction on n; $n = 1$ is clear. Take an eigenvalue λ_1 (T7).
2. Splitting result for $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$, $U = \operatorname{range}(T - \lambda_1 I)^n$. U is invariant (T5), and $\dim U < n$ since $G(\lambda_1, T) \neq 0$ (Prop. 71).
3. Eigenvalues of $T|_U$ are eigenvalues of T, but not λ_1 ($T - \lambda_1 I$ is injective on U). By induction $U = \bigoplus_{i \geq 2} G(\lambda_i, T|_U)$, and $G(\lambda_i, T|_U) \subseteq G(\lambda_i, T)$.
4. So the $G(\lambda_i, T)$ span V. Direct: if $u_1 + \cdots + u_m = 0$, $u_i \in G(\lambda_i, T)$, the nonzero u_i would be independent (Prop. 72); so all are 0.

□

On each $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with N_i nilpotent. So everything reduces to nilpotent operators (Ch. 15).

Definition (Def. 38)**. Algebraic multiplicity** of λ: $\dim G(\lambda, T)$. **Geometric multiplicity**: $\dim E(\lambda, T)$.

- geometric $\leq$ algebraic, since $E \subseteq G$.
- Over $\mathbb{C}$, the algebraic multiplicities add up to n (Prop. 73).
- Over $\mathbb{C}$, T is diagonalizable $\iff$ geometric = algebraic for every eigenvalue.
  Why: The sum $\sum E(\lambda_i, T)$ is direct (T7), and T is diagonalizable iff it is V, i.e. $\sum \dim E = n = \sum \dim G$. Since each $\dim E \leq \dim G$, the sums agree iff every term agrees.

Theorem (Prop. 74 / LADR 8.26, 8.29: Cayley–Hamilton)
Over $\mathbb{C}$, let $d_i = \dim G(\lambda_i, T)$ and $\chi_T(x) = \prod_i (x - \lambda_i)^{d_i}$ (the **characteristic polynomial**). Then $\chi_T(T) = 0$. So $\deg \chi_T = n$, $m_T \mid \chi_T$, and $\deg m_T \leq n$.

How to find it

- To kill T, it is enough to kill each $G(\lambda_i, T)$, since they span V.
- On $G(\lambda_i, T)$, $T - \lambda_i I$ is nilpotent on a d_i-dimensional space, so its d_i-th power is 0 there (Prop. 70 inside that space).
- Multiply these factors; they commute, so the right factor can act first.

Proof. 1. $N = (T - \lambda_i I)|_{G(\lambda_i, T)}$ has $G(0, N) = G(\lambda_i, T)$, so $N^{d_i} = 0$ by Prop. 70 in that space.
2. For $v \in G(\lambda_i, T)$: $\chi_T(T)v = \prod_{j \neq i} (T - \lambda_j I)^{d_j} (T - \lambda_i I)^{d_i} v = 0$ (T5).
3. The G's span V, so $\chi_T(T) = 0$; then $m_T \mid \chi_T$ (T9) and $\deg \chi_T = \sum d_i = n$.

□

- For a matrix, $\chi_A(x) = \det(xI - A)$ (stated in the notes; needs determinants).
- 2×2: $\chi_A = x^2 - (\operatorname{tr} A)x + \det A$.
  Why: Multiplying out gives $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \neq cI$, m_A has degree 2 and divides both this polynomial and χ_A, so all three agree. If $A = cI$, both are $(x - c)^2$.
- χ_T and m_T have the same roots; the exponents in m_T can be smaller.

Proposition (Diagonal of a triangular matrix)
Over $\mathbb{C}$, if $\mathcal{M}(T)$ is upper-triangular, then $\dim G(\lambda, T)$ is the number of times λ appears on the diagonal.

How to find it

- Show $\leq$ for each λ by counting rank. Both sides add up to n, so no inequality can be strict.

Proof. 1. Let c_λ be the count. $\mathcal{M}((T-\lambda I)^n)$ is triangular with diagonal $(a_{jj}-\lambda)^n$, nonzero in $n-c_\lambda$ places.
2. Those columns are independent (T8), so rank $\geq n-c_\lambda$ and $\dim G(\lambda,T)\leq c_\lambda$ (Prop. 70, T1).
3. $n=\sum_\lambda \dim G(\lambda,T)\leq \sum_\lambda c_\lambda = n$ (Prop. 73), so every inequality is an equality.

□

Example (Computing $G(\lambda, A)$ (Ex. 61))
$A=\left(\begin{smallmatrix}1&1&1\\0&1&1\\0&0&3\end{smallmatrix}\right)$, $\lambda=1$.

1. $\operatorname{null}(A-I)=\operatorname{span}(e_1)$, $\operatorname{null}(A-I)^2=\operatorname{span}(e_1,e_2)$, $\operatorname{null}(A-I)^3=\operatorname{span}(e_1,e_2)$.
2. Two consecutive null spaces agree, so the chain has stopped: $G(1,A)=\operatorname{span}(e_1,e_2)$.
3. Shortcut: 1 appears twice on the diagonal, so you can stop at dimension 2.

Traps

- $G(\lambda,T)$ can be strictly bigger than $E(\lambda,T)$. Over $\mathbb{C}$ they agree for all λ exactly when T is diagonalizable.
- Prop. 73 needs eigenvalues: true over $\mathbb{C}$; over $\mathbb{R}$ only when m_T splits.
- $V=\operatorname{null}T\oplus\operatorname{range}T$ can fail: for $N=\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$, $\operatorname{null}N=\operatorname{range}N=\operatorname{span}(e_1)$. The splitting needs the power n.
- The power n always suffices; the chain often stops sooner.

§1.3 Nilpotent operators and Jordan form (Ch. 15)

The question. A complex operator is a direct sum of pieces $\lambda I+N$ with N nilpotent (Prop. 73), and λI looks the same in every basis. So we only need a good basis for a nilpotent N.

- Simplest nilpotent: a shift $v\mapsto Nv\mapsto\cdots\mapsto N^dv\mapsto 0$. In the basis $N^dv,\ldots,Nv,v$ its matrix has 1s just above the diagonal.
- The hope: every nilpotent operator is a direct sum of shifts.
- A basis of generalized eigenvectors is not enough: e_1,e_2,e_3 are generalized eigenvectors for 1 of I, $B=\left(\begin{smallmatrix}3&-1&-1\\2&0&-1\\2&-1&0\end{smallmatrix}\right)$ and $C=\left(\begin{smallmatrix}3&-2&0\\1&1&-1\\0&2&-1\end{smallmatrix}\right)$, yet $\dim E(1)=3,2,1$, so none are similar.

Definition (Def. 39)**.** N is **nilpotent** if $N^k=0$ for some k; the least such k is the **nilpotency index**.

- $N^n=0$: $V=G(0,N)=\operatorname{null}N^n$ (Prop. 70). So $m_N=x^k$, $k\leq n$.
- The only eigenvalue is 0 (if $V\neq 0$): $Nv=\lambda v\Rightarrow 0=N^kv=\lambda^kv$. Over $\mathbb{C}$, conversely, only eigenvalue $0\Rightarrow V=G(0,N)$ (Prop. 73).
- $\operatorname{tr}N=0$: the matrix of Prop. 75 has zero diagonal.
- $N\neq 0$ is not diagonalizable: a diagonal matrix with all eigenvalues 0 is 0.
- Index $=\deg m_N=$ size of the largest Jordan block $\leq n$. HW8 P1: D on $\mathcal{P}_n(\mathbb{C})$ has index $n+1$, one block $J_{n+1}(0)$ in the basis $1,x,\frac{x^2}{2!},\ldots,\frac{x^n}{n!}$.
- $(I-N)^{-1}=I+N+\cdots+N^{n-1}$: the product telescopes to $I-N^n=I$.

Proposition (Chain lemma)
If $N^dv\neq 0$ and $N^{d+1}v=0$, then $v,Nv,\ldots,N^dv$ are independent.

How to find it

- Apply a power of N that kills every term but one: N^d leaves only $c_0 N^d v$.

Proof. Apply N^d to $\sum_j c_j N^j v = 0$: get $c_0 N^d v = 0$, so $c_0 = 0$. Then apply N^{d-1} to get $c_1 = 0$, and so on. □

Theorem (Prop. 75 / LADR 8.45: nilpotent normal form)
A nilpotent N has a basis made of chains $u, Nu, \ldots, N^k u$ ($N^{k+1}u = 0$). Listing each chain as $N^k u, \ldots, u$, $\mathcal{M}(N)$ has only 1s and 0s on the superdiagonal and 0 elsewhere.

How to find it

- Induct on $\dim V$. N is not injective, so not surjective (T2): $U = \operatorname{range} N$ is smaller and invariant, and by induction has a chain basis.
- Lengthen: each chain in U starts at some $v_i \in \operatorname{range} N$, so $v_i = Nu_i$. Put u_i on top.
- Fill in: the chains end in $\operatorname{null} N \cap \operatorname{range} N$. Vectors of $\operatorname{null} N$ outside $\operatorname{range} N$ are missing; add them as chains of length 1.
- If the total count is n, independence is enough (T3).

Proof. 1. If $N = 0$, any basis works. Otherwise $0 < \dim U < n$, and U has a chain basis $v_i, Nv_i, \ldots, N^{d_i} v_i$ ($i \le r$), with $N^{d_i+1} v_i = 0$.

2. *The bottoms $N^{d_i} v_i$ are a basis of* $\operatorname{null} N \cap U$. They are independent and in $\operatorname{null} N$. If $x = \sum c_{ij} N^j v_i$ and $Nx = 0$, then $0 = \sum_{j<d_i} c_{ij} N^{j+1} v_i$, a combination of distinct basis vectors of U, so $c_{ij} = 0$ for $j < d_i$.
3. Choose u_i with $Nu_i = v_i$. Extend the bottoms by $w_1, \ldots, w_l$ to a basis of $\operatorname{null} N$, so $\dim \operatorname{null} N = r + l$.
4. New list: $u_i, Nu_i, \ldots, N^{d_i+1} u_i$ and $w_1, \ldots, w_l$. Length: $\sum (d_i + 2) + l = \dim U + r + l = \dim \operatorname{range} N + \dim \operatorname{null} N = n$.
5. *Independent.* Apply N to a relation: the w's and the bottoms die, leaving a relation among the basis of U, so those coefficients are 0. What remains is a relation among the basis of $\operatorname{null} N$, so it is trivial.

□

Definition (Def. 40)**. Jordan block**: $J_k(\lambda) = \lambda I + N_k$ ($k \times k$, 1s on the superdiagonal), so $Je_1 = \lambda e_1$, $Je_j = \lambda e_j + e_{j-1}$. **Jordan normal form**: block diagonal with Jordan blocks. **Jordan basis**: a basis giving Jordan form.

Theorem (Prop. 76 / LADR 8.46: Jordan normal form)
Every operator on a finite-dimensional complex space has a Jordan basis.

How to find it

- Combine the two reductions: split into generalized eigenspaces (Prop. 73), then apply Prop. 75 to the nilpotent part on each.

Proof. On each invariant $G(\lambda_i, T)$, $N_i = (T - \lambda_i I)|_G$ is nilpotent. Prop. 75 gives a basis where N_i is a sum of blocks N_k, so $T = \lambda_i I + N_i$ is a sum of blocks $J_k(\lambda_i)$. Concatenate the bases (T10). □

Reading the Jordan form. For each eigenvalue λ:

- number of blocks for λ = geometric multiplicity $\dim E(\lambda, T)$;
- total size of those blocks = algebraic multiplicity $\dim G(\lambda, T)$;
- largest block for λ = exponent of $(x-\lambda)$ in m_T;
- $\chi_T = \prod (x-\lambda)^{\text{alg. mult.}}$, $m_T = \prod (x-\lambda)^{\text{largest block}}$;
- diagonalizable $\iff$ all blocks 1×1 $\iff$ m_T has no repeated root.

Why: One block $J_k(\lambda)$: N_k^j shifts by j places, so $\dim \operatorname{null} N_k^j = \min(j,k)$. Hence $E = \operatorname{span}(e_1)$, G is everything, and $m = \chi = (x-\lambda)^k$. For a direct sum, null spaces add (T10) and m is the lcm. A block $J_j(\mu)$, $\mu \neq \lambda$, adds nothing to $\operatorname{null}(T-\lambda I)^k$: if $((\mu-\lambda)I + N_j)v = 0$ then $N_j v = (\lambda-\mu)v$, and a nilpotent has only the eigenvalue 0.

Theorem (Prop. 77: counting blocks)

Over $\mathbb{C}$, let $d_k = \dim \operatorname{null}(T-\lambda I)^k$ ($d_0 = 0$). For $k \geq 1$:

- number of blocks for λ of size $\geq k = d_k - d_{k-1}$;
- number of size exactly $k = 2d_k - d_{k-1} - d_{k+1}$.

So the Jordan form is unique up to order, and two complex matrices are similar iff they have the same Jordan form.

How to find it

- The d_k come from T alone, with no basis. If block counts are expressed through them, the blocks cannot depend on the basis.
- Null spaces of block diagonal matrices add, so count block by block.
- A block $J_j(\lambda)$ contributes $\min(k,j)$, which grows from $k-1$ to k exactly when $j \geq k$.

Proof. 1. In a Jordan basis, $d_k = \sum_{\text{blocks}} \dim \operatorname{null}(J-\lambda I)^k$. Blocks for $\mu \neq \lambda$ contribute 0; $J_j(\lambda)$ contributes $\min(k,j)$.

2. $\min(k,j) - \min(k-1,j)$ is 1 if $j \geq k$, else 0. Summing: $d_k - d_{k-1} = \#\{\text{blocks with } j \geq k\}$. Exactly k: $(d_k - d_{k-1}) - (d_{k+1} - d_k)$.

3. Similar matrices are one operator in two bases (T11), so they have the same d_k and the same Jordan form. Conversely, matrices with the same Jordan form are both similar to it.

□

Example (Jordan form from null-space dimensions (Ex. 63))

$T \in \mathcal{L}(\mathbb{C}^8)$ with eigenvalues $1, 2$. Method: list $d_0, d_1, \ldots$, take differences, then differences of differences.

λ	d_0	d_1	d_2	d_3	d_4	blocks
1	0	2	4	4		sizes ≥ 1: 2, ≥ 2: 2, ≥ 3: 0 $\Rightarrow J_2(1), J_2(1)$
2	0	2	3	4	4	sizes ≥ 1: 2, ≥ 2: 1, ≥ 3: 1, ≥ 4: 0 $\Rightarrow J_3(2), J_1(2)$

($d_4 = 4$ for $\lambda = 2$ because $\dim G(2,T) = 8-4$.) Answer: $J_2(1) \oplus J_2(1) \oplus J_3(2) \oplus J_1(2)$.

Example (Finding a Jordan basis (Ex. 62))

$A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$: only eigenvalue 1, $d_1, d_2, d_3 = 1, 2, 3$, so one block $J_3(1)$.

1. Pick the top: $v_3 \in \operatorname{null}(A-I)^3$, $v_3 \notin \operatorname{null}(A-I)^2$. Here $v_3 = e_3$.
2. Go down: $v_2 = (A-I)v_3 = (3,2,0)$, $v_1 = (A-I)v_2 = (4,0,0)$ (an eigenvector).
3. Check: $Av_1 = v_1$, $Av_2 = v_2 + v_1$, $Av_3 = v_3 + v_2$. That is $J_3(1)$.

Why start at the top: the top vector determines the whole chain, and a top outside $\operatorname{null}(A-\lambda I)^{k-1}$ guarantees full length k (chain lemma).

Several blocks: get the block sizes first (Prop. 77). For each block of size j, largest first, pick a top $v \in \operatorname{null}(A-\lambda I)^j \setminus \operatorname{null}(A-\lambda I)^{j-1}$ and keep its chain only if its bottom $(A-\lambda I)^{j-1}v$ is independent of the bottoms already chosen. (Independent of the vectors so far is not enough: in $J_3(0) \oplus J_2(0)$ with chains e_1, e_2, e_3 and f_1, f_2, the top $f_1 + e_2$ gives a chain ending at e_1 again.)

Powers. $J = \lambda I + N$ with λI and N commuting, so the binomial theorem applies and $N^j = 0$ for $j \geq k$:

$$J_k(\lambda)^m = \sum_{j=0}^{\min(m,k-1)} \binom{m}{j} \lambda^{m-j} N^j, \qquad J_3(\lambda)^m = \begin{pmatrix} \lambda^m & m\lambda^{m-1} & \binom{m}{2}\lambda^{m-2} \\ 0 & \lambda^m & m\lambda^{m-1} \\ 0 & 0 & \lambda^m \end{pmatrix}.$$

- Example: $J_3(1)^{100}$ has 100 and 4950 above the diagonal.
- Any Jordan matrix is $J = D + N$, D diagonal, N nilpotent, $DN = ND$ (on each block $D = \lambda I$); this is the Jordan–Chevalley decomposition. If J is invertible, also $J = DU$ with $U = I + D^{-1}N$, $DU = UD$.
- **Square roots (LADR 8.39–8.41).** $I + N$ has a square root: choose a_j with $(I + a_1 N + \cdots + a_{n-1}N^{n-1})^2 = I + N$ by matching coefficients ($2a_1 = 1$, $2a_2 + a_1^2 = 0$, …); powers N^n and up vanish. This is the series $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$ cut off. Over $\mathbb{C}$, every invertible operator has a square root (apply this to $\lambda_i(I + N_i/\lambda_i)$ on each $G(\lambda_i, T)$).

Traps

- χ_T and m_T do not fix the Jordan form: $J_2(0) \oplus J_2(0)$ and $J_2(0) \oplus J_1(0) \oplus J_1(0)$ both have $\chi = x^4$, $m = x^2$.
- Adding the number of blocks is still not enough: $J_3 \oplus J_3 \oplus J_1$ and $J_3 \oplus J_2 \oplus J_2$ (all for 0) agree in χ, m and block count, but $d_2 = 5$ vs. 6. You need all the d_k.
- Order each chain from the bottom (the eigenvector) up; reversed, the 1s go below the diagonal.
- Jordan form needs $\mathbb{C}$ (or split m_T): the rotation of $\mathbb{R}^2$ has none over $\mathbb{R}$.

§1.4 Trace (Ch. 16)

The question. The matrix of an operator depends on the basis. Which numbers computed from it do not? The trace (sum of the diagonal) is one, and it equals the sum of the eigenvalues. It gives information about eigenvalues without computing them.

Definition (Def. 41, 42 / LADR 8.47, 8.50, 8.51)**.** $\operatorname{tr} A = a_{11} + \cdots + a_{nn}$. For $T \in \mathcal{L}(V)$, $\operatorname{tr} T = \operatorname{tr} \mathcal{M}(T)$ in any basis.

Proposition (Prop. 78 / LADR 8.49)

For A $(m \times n)$ and B $(n \times m)$, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Hence $\operatorname{tr}(S^{-1}AS) = \operatorname{tr} A$, and $\operatorname{tr} T$ is well defined.

How to find it

- Write both traces as double sums; they are the same sum in a different order.
- Similarity: group $S^{-1}AS$ as $(S^{-1}A)S$ and move S to the front.

Proof. 1. $\operatorname{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji} = \sum_j \sum_i b_{ji} a_{ij} = \operatorname{tr}(BA)$.
2. $\operatorname{tr}\big((S^{-1}A)S\big) = \operatorname{tr}\big(S(S^{-1}A)\big) = \operatorname{tr} A$. Matrices of one operator are similar (T11).

□

Proposition (Prop. 79 / LADR 8.52)
Over $\mathbb{C}$, $\operatorname{tr} T = \sum_i d_i \lambda_i$ over the distinct eigenvalues, $d_i = \dim G(\lambda_i, T)$. Also $-\operatorname{tr} T$ is the coefficient of x^{n-1} in χ_T.

How to find it

- Trace can be computed in any basis, so use the one with the simplest diagonal: a Jordan basis lists each λ_i exactly d_i times.

Proof. In a Jordan basis the blocks for λ_i have total size d_i. Expanding $\prod (x - \lambda_i)^{d_i}$, the x^{n-1} coefficient is $-\sum d_i \lambda_i$. □

Proposition (Prop. 80)
$\operatorname{tr}(ST - TS) = 0$. So on $V \neq \{0\}$ finite-dimensional, there are no S, T with $ST - TS = I$.

How to find it

- To show an equation has no solution, apply a linear function that gives the two sides different values.
- Trace ignores the order of a product, so it sends every $ST - TS$ to 0; but $\operatorname{tr} I = \dim V \neq 0$.

Proof. $\operatorname{tr}(ST - TS) = \operatorname{tr}(ST) - \operatorname{tr}(TS) = 0$ (linearity, Prop. 78), while $\operatorname{tr} I = \dim V > 0$. □

- **Linear:** $\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B$, $\operatorname{tr}(cA) = c \operatorname{tr} A$ (diagonals add).
- **Cyclic:** $\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$ (Prop. 78 with A and BC). Not every order: $A = E_{12}, B = E_{21}, C = E_{11}$ give $\operatorname{tr}(ABC) = 1$, $\operatorname{tr}(BAC) = 0$.
- **Projections:** $P^2 = P \Rightarrow \operatorname{tr} P = \operatorname{rank} P$. In a basis for $\operatorname{null} P \oplus \operatorname{range} P$ (Prop. 91), $\mathcal{M}(P) = \operatorname{diag}(0, \ldots, 0, 1, \ldots, 1)$.
- **Nilpotent:** $\operatorname{tr} N = 0$, and $\operatorname{tr} N^k = 0$. The converse fails: $\operatorname{diag}(1, -1)$.
- **Polynomials:** over $\mathbb{C}$, $\operatorname{tr} p(T) = \sum_i d_i p(\lambda_i)$ ($p(T)$ is triangular in a Jordan basis with diagonal $p(\lambda_i)$).
- 2×2**:** $\lambda_1 + \lambda_2 = \operatorname{tr} A$, $\lambda_1 \lambda_2 = \det A$. Ex. 64: $\operatorname{tr}\left(\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{smallmatrix}\right) = 12$ is the sum of eigenvalues that are hard to find.
- **Detecting** 0**:** if $\operatorname{tr}(ST) = 0$ for all S, then $T = 0$ (take $S = E_{ji}$: $\operatorname{tr}(E_{ji}\mathcal{M}(T)) = t_{ij}$).

Traps

- $\operatorname{tr}(AB) \neq \operatorname{tr} A \cdot \operatorname{tr} B$ ($A = B = I_2$: $2 \neq 4$).
- $\operatorname{tr} T = 0$ does not make T nilpotent.
- $ST - TS = I$ is possible in infinite dimensions: on $\mathcal{P}(\mathbb{R})$, $D(xp) - xD(p) = p$.

§1.5 Inner products and norms (Ch. 17)

The question. So far vector spaces have no lengths or angles. In $\mathbb{R}^n$ the dot product gives both. We want them in any vector space (polynomials, functions), so we take the properties of the dot product that proofs use as the definition.

$\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ (over $\mathbb{R}$ ignore the bars). From here on V may be infinite-dimensional; finite dimension is stated when needed.

Definition (Def. 43 / LADR 6.2)**.** An **inner product** $\langle -, - \rangle : V \times V \to \mathbb{F}$ satisfies

(i) positivity: $\langle v, v \rangle > 0$ for $v \neq 0$;
(ii) linearity in the first slot: $\langle \lambda u + v, w \rangle = \lambda \langle u, w \rangle + \langle v, w \rangle$;
(iii) conjugate symmetry: $\langle u, v \rangle = \overline{\langle v, u \rangle}$.

- **Why the conjugate:** without it, $(i, 1) \cdot (i, 1) = i^2 + 1 = 0$, a nonzero vector of length 0. With $z\bar{z} = |z|^2 \geq 0$ lengths are positive, and symmetry must become conjugate symmetry.
- $\langle 0, v \rangle = 0$: $\langle 0, v \rangle = \langle 0 + 0, v \rangle = 2\langle 0, v \rangle$.
- Second slot: $\langle u, \lambda v + w \rangle = \overline{\langle \lambda v + w, u \rangle} = \bar{\lambda} \langle u, v \rangle + \langle u, w \rangle$ (conjugate-linear).
- $\langle v, v \rangle$ is real (it equals its own conjugate).
- **Examples:** dot product $\sum w_j \bar{z}_j$ on $\mathbb{F}^n$ (Ex. 65); L^2 product $\int_a^b f\bar{g}$ on continuous functions (Ex. 66); weighted $x_1 y_1 + 2x_2 y_2$.
- **Checking a candidate:** (ii) and (iii) are usually immediate. Test positivity: can $\langle v, v \rangle \leq 0$ for some $v \neq 0$? For integrals: $\int p^2 = 0$ forces the continuous $p^2 \geq 0$ to vanish on the interval, so p has infinitely many roots and $p = 0$.

Definition (Def. 44 / Prop. 81 / LADR 6.7, 6.17)**.** A **norm**: $\|v\| > 0$ for $v \neq 0$; $\|\lambda v\| = |\lambda| \|v\|$; $\|u + v\| \leq \|u\| + \|v\|$. An inner product gives $\|v\| = \sqrt{\langle v, v \rangle}$.

- Homogeneity: $\|\lambda v\|^2 = \lambda \bar{\lambda} \langle v, v \rangle = |\lambda|^2 \|v\|^2$. The triangle inequality needs Cauchy–Schwarz.
- **Pythagoras:** if $\langle u, v \rangle = 0$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$ (expand; the cross terms are 0).
- The converse holds over $\mathbb{R}$ but not over $\mathbb{C}$ (the cross term is $2\,\mathrm{Re}\langle u, v \rangle$): in $\mathbb{C}$, $u = 1$, $v = i$ give $|1 + i|^2 = 1 + 1$ but $\langle u, v \rangle = -i$.

Proposition (LADR 6.13: orthogonal decomposition)

If $v \neq 0$: $u = cv + w$ with $w \perp v$, where $c = \dfrac{\langle u, v \rangle}{\|v\|^2}$. So the projection onto a line is $P_{\mathrm{span}(v)} u = \dfrac{\langle u, v \rangle}{\|v\|^2} v$.

How to find it

- The most useful idea in Ch. 17–18: the *shadow* of u on the line through v.
- Want c with $u - cv \perp v$. Solve $0 = \langle u - cv, v \rangle = \langle u, v \rangle - c\|v\|^2$.
- Cauchy–Schwarz, Gram–Schmidt, projections and best approximation all reuse this.

Theorem (Prop. 82 / LADR 6.14: Cauchy–Schwarz)

$|\langle u, v \rangle| \leq \|u\| \|v\|$, with equality iff one of u, v is a multiple of the other.

How to find it

- A vector is at least as long as its shadow on any line.
- Decompose $u = cv + w$; Pythagoras gives $\|u\|^2 \geq |c|^2 \|v\|^2$, and $|c| \|v\| = |\langle u, v \rangle| / \|v\|$.
- Equality means nothing was thrown away: $w = 0$. (The notes expand $\langle z, z \rangle \geq 0$ with $z = -w$; same computation.)

Proof. 1. If $v = 0$, both sides are 0 and $v = 0 \cdot u$.

2. If $v \neq 0$, write $u = cv + w$, $c = \langle u, v \rangle / \|v\|^2$, $w \perp v$. Pythagoras: $\|u\|^2 = |c|^2 \|v\|^2 + \|w\|^2 \geq \dfrac{|\langle u, v \rangle|^2}{\|v\|^2}$.

3. Equality iff $w = 0$, i.e. $u = cv$. Conversely $u = av$ gives $|\langle u, v \rangle| = |a| \|v\|^2 = \|u\| \|v\|$.

□

Proposition (Prop. 81(iii) / LADR 6.17: triangle inequality)
$\|u+v\| \le \|u\| + \|v\|$, with equality iff one is a nonnegative multiple of the other.

How to find it

- Compare squares. The cross term $2\,\mathrm{Re}\langle u, v\rangle$ is at most $2|\langle u, v\rangle|$, which Cauchy–Schwarz bounds.

Proof. 1. $\|u+v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2 \le \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2 \le (\|u\| + \|v\|)^2$.
2. Equality needs $\mathrm{Re}\langle u, v\rangle = |\langle u, v\rangle| = \|u\|\|v\|$: then $u = av$ (or $v = 0$), and $\langle u, v\rangle = a\|v\|^2 \ge 0$ forces $a \ge 0$.
3. Conversely, $u = av$ with $a \ge 0$ gives $\|u+v\| = (1+a)\|v\| = \|u\| + \|v\|$.

□

Proposition (Prop. 83 / LADR 6.21, 6A.28: parallelogram law, polarization)
A norm comes from an inner product iff $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$. Then

$$\langle u, v\rangle = \tfrac{1}{4}\big(\|u+v\|^2 - \|u-v\|^2\big)\ (\mathbb{R}), \qquad \langle u, v\rangle = \tfrac{1}{4}\big(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\big)\ (\mathbb{C}).$$

How to find it

- Expand both squares: the cross terms have opposite signs. Adding cancels them (the law); subtracting isolates them (polarization).
- Over $\mathbb{C}$ subtracting only gives $\mathrm{Re}\langle u, v\rangle$; repeat with iv for the imaginary part.
- The converse (the law forces an inner product) is not proved in the notes.

Proof. 1. $\|u \pm v\|^2 = \|u\|^2 \pm 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2$. Add: the law. Subtract: $4\,\mathrm{Re}\langle u, v\rangle$.
2. $\mathrm{Re}\langle u, iv\rangle = \mathrm{Re}(-i\langle u, v\rangle) = \mathrm{Im}\langle u, v\rangle$, so $\|u+iv\|^2 - \|u-iv\|^2 = 4\,\mathrm{Im}\langle u, v\rangle$.

□

Use: the max norm $\max(|x|, |y|)$ on $\mathbb{R}^2$ fails the law ($u = e_1, v = e_2$: $1+1 \ne 2+2$), so it comes from no inner product.

- **Angle** (Def. 45 / LADR 6A.16, real spaces): for $u, v \ne 0$, $\cos\theta = \frac{\langle u, v\rangle}{\|u\|\|v\|}$, $\theta \in [0, \pi]$. Cauchy–Schwarz puts the quotient in $[-1, 1]$. Ex. 68: $(1, 0), (1, 1)$ make angle $\pi/4$.
- **Using Cauchy–Schwarz:** write the quantity as $\langle u, v\rangle$ with a clever second vector, usually all 1s or the constant 1.
  - $(a_1 + \cdots + a_n)^2 \le n\sum a_i^2$, equality iff all a_i equal.
  - $\big(\int_0^1 f\big)^2 \le \int_0^1 f^2$; $\int_0^\pi \sqrt{\sin x}\,dx \le \sqrt{2\pi}$ (Ex. 67).
  - Weighted: $\big(\sum a_k b_k\big)^2 \le \big(\sum k a_k^2\big)\big(\sum b_k^2/k\big)$, writing $a_k b_k = (\sqrt{k}\,a_k)(b_k/\sqrt{k})$.
- **Reverse triangle inequality:** $\big|\|u\| - \|v\|\big| \le \|u - v\|$, since $\|u\| \le \|u - v\| + \|v\|$ and symmetrically.

Traps

- Over $\mathbb{C}$, scalars leave the second slot conjugated.
- $x_1y_1 - x_2y_2$ and $(x_1 + x_2)(y_1 + y_2)$ are not inner products: $\langle e_2, e_2\rangle = -1$, $\langle (1, -1), (1, -1)\rangle = 0$.
- Cauchy–Schwarz equality: "dependent". Triangle equality: "nonnegative multiple".

§1.6 Orthogonality (Ch. 18)

The question. In a general basis, finding coordinates means solving a linear system. If the basis vectors are perpendicular unit vectors, each coordinate is one inner product. We want such bases, a way to build them, and

to use them to split spaces into perpendicular pieces.

Definition (Def. 46)**.** $u \perp v$ if $\langle u, v\rangle = 0$. $e_1, \dots, e_m$ is **orthonormal** if $\langle e_i, e_j\rangle = \delta_{ij}$.

Proposition (Prop. 84–86 / LADR 6.24, 6.25, 6.30)

For an orthonormal list: $\|\sum a_i e_i\|^2 = \sum |a_i|^2$, and the list is independent. For an orthonormal basis:

$$v = \sum_i \langle v, e_i\rangle e_i, \qquad \|v\|^2 = \sum_i |\langle v, e_i\rangle|^2.$$

How to find it

- Norm: Pythagoras, one term at a time.
- Independence: a combination equal to 0 has norm 0.
- Coordinates: take the inner product with e_k; every other term dies.

Proof. 1. $e_m \perp a_1 e_1 + \cdots + a_{m-1} e_{m-1}$, so Pythagoras adds $|a_m|^2$; induct.
2. $\sum a_i e_i = 0 \Rightarrow \sum |a_i|^2 = 0 \Rightarrow$ all $a_i = 0$.
3. If $v = \sum a_i e_i$, then $\langle v, e_k\rangle = \sum a_i \langle e_i, e_k\rangle = a_k$.

□

Parseval (infinite-dimensional version): for $f(x) = \sum c_n e^{2\pi i n x}$ on $[0,1]$, $\sum |c_n|^2 = \int_0^1 |f|^2$. With $f(x) = x = \frac{1}{2} - \frac{1}{2\pi i}\sum_{n\neq 0} \frac{1}{n} e^{2\pi i n x}$ this gives $\frac{1}{4} + \frac{1}{4\pi^2}\sum_{n\neq 0} \frac{1}{n^2} = \frac{1}{3}$, so $\sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Theorem (Prop. 87 / LADR 6.32: Gram–Schmidt)

For independent $v_1, \dots, v_n$: $e_1 = v_1/\|v_1\|$, and for $j \geq 2$

$$w_j = v_j - \sum_{k<j} \langle v_j, e_k\rangle e_k, \qquad e_j = w_j/\|w_j\|.$$

Then $e_1, \dots, e_n$ is orthonormal and $\operatorname{span}(e_1, \dots, e_j) = \operatorname{span}(v_1, \dots, v_j)$ for every j.

How to find it

- Build one vector at a time. We need a new unit vector perpendicular to $e_1, \dots, e_{j-1}$ that brings in v_j.
- Remove from v_j its shadow on each e_k (just $\langle v_j, e_k\rangle e_k$, since $\|e_k\| = 1$). Then scale.
- The only risk is $w_j = 0$; independence rules it out.

Proof. 1. If $w_j = 0$, then $v_j \in \operatorname{span}(e_1, \dots, e_{j-1}) = \operatorname{span}(v_1, \dots, v_{j-1})$, contradicting independence.
2. For $k < j$: $\langle w_j, e_k\rangle = \langle v_j, e_k\rangle - \langle v_j, e_k\rangle = 0$. And $\|e_j\| = 1$.
3. $e_j \in \operatorname{span}(v_1, \dots, v_j)$, so the spans are nested; both have dimension j, so they are equal.

□

- Every finite-dimensional inner product space has an orthonormal basis; every orthonormal list extends to one (extend to a basis, then Gram–Schmidt; the first vectors do not change) (LADR 6.35, 6.36).
- An orthonormal list of length $\dim V$ is an orthonormal basis (Prop. 85, T3).
- If the v_j are dependent, $w_j = 0$ exactly when $v_j \in \operatorname{span}(v_1, \dots, v_{j-1})$: skip it and continue.
- **Schur (LADR 6.38):** over $\mathbb{C}$, T is upper-triangular in some *orthonormal* basis. Take a triangular basis (T8) and apply Gram–Schmidt: it keeps the spans $\operatorname{span}(v_1, \dots, v_j)$, which are invariant, so the matrix stays triangular.

Example (Ex. 70: Gram–Schmidt on $\mathcal{P}_2$, $\langle p, q\rangle = \int_0^1 p\bar{q}$) 1. $\|1\| = 1$: $e_1 = 1$.

2. $w_2 = x - \langle x, 1\rangle = x - \frac{1}{2}$, $\|w_2\|^2 = \frac{1}{12}$: $e_2 = \sqrt{3}(2x-1)$.
3. $w_3 = x^2 - \frac{1}{3} - \frac{1}{2}(2x-1) = x^2 - x + \frac{1}{6}$, $\|w_3\| = \frac{1}{6\sqrt{5}}$: $e_3 = \sqrt{5}(6x^2 - 6x + 1)$.

Order of work: w_j, then $\|w_j\|^2$, then divide. Check $w_j \perp$ the earlier e's. (These are the shifted Legendre polynomials; on $[-1, 1]$, HW10 P1, the Legendre polynomials.)

Proposition (Prop. 88 / LADR 6.42, 6.58: Riesz representation)
For finite-dimensional V, every $\psi \in V'$ is $\psi(u) = \langle u, v\rangle$ for a unique v. In an orthonormal basis, $v = \sum_j \overline{\psi(e_j)} e_j$.

How to find it

- On $\mathbb{R}^n$ a functional is a row vector a and $\psi(u) = a \cdot u$: the representing vector is a.
- In general, write $v = \sum a_j e_j$ and test on basis vectors: $\psi(e_i) = \langle e_i, v\rangle = \bar{a}_i$. That forces the coefficients.

Proof. 1. With $v = \sum \overline{\psi(e_j)} e_j$: $\langle e_i, v\rangle = \psi(e_i)$. Two linear functionals equal on a basis are equal.

2. Unique: if $\langle u, v\rangle = \langle u, v'\rangle$ for all u, take $u = v - v'$ to get $\|v - v'\|^2 = 0$.

(The notes use $\Gamma(v) = \langle -, v\rangle$: real-linear, not complex-linear since $\Gamma(iv) = -i\Gamma(v)$, injective, and $\dim_{\mathbb{R}} V = \dim_{\mathbb{R}} V'$.) □

Definition (Def. 47 / LADR 6.46, 6.47). $U^\perp = \{v : \langle v, u\rangle = 0 \text{ for all } u \in U\}$, a subspace (linearity in the first slot).

- $\{0\}^\perp = V$; $V^\perp = \{0\}$; $U \cap U^\perp = \{0\}$ (a vector perpendicular to itself is 0).
- $U \subseteq W \Rightarrow W^\perp \subseteq U^\perp$.
- $(U + W)^\perp = U^\perp \cap W^\perp$ (perpendicular to all $u + w$ iff to all u and all w).
- $(U \cap W)^\perp = U^\perp + W^\perp$ (finite dim., HW10 P4): take $\perp$ of $(U^\perp + W^\perp)^\perp = U \cap W$ (Prop. 90).
- Finite dim.: $U^\perp = \{0\} \iff U = V$; $U^\perp = V \iff U = \{0\}$ (Prop. 89).
- **Computing $U^\perp$:** require $\langle v, u_i\rangle = 0$ for a spanning list of U and solve the linear system. Under Riesz, $U^\perp$ matches the annihilator U^0.

Theorem (Prop. 89 / LADR 6.49, 6.50)
If U is finite-dimensional, $V = U \oplus U^\perp$. If V is finite-dimensional, $\dim U^\perp = \dim V - \dim U$.

How to find it

- Direct: $U \cap U^\perp = 0$.
- Sum: split v into a piece in U and a piece $\perp U$. For one vector that was the orthogonal decomposition; for a subspace, remove the shadow on each vector of an orthonormal basis of U.

Proof. 1. Direct: if $u \in U \cap U^\perp$, then $\langle u, u\rangle = 0$, so $u = 0$.

2. Take an orthonormal basis $e_1, \ldots, e_m$ of U (Gram–Schmidt). Set $u = \sum \langle v, e_k\rangle e_k \in U$, $w = v - u$.
3. $\langle w, e_k\rangle = \langle v, e_k\rangle - \langle v, e_k\rangle = 0$ for each k, so $w \perp U$, and $v = u + w$.

□

Proposition (Prop. 90 / LADR 6.52)
If U is finite-dimensional, $(U^\perp)^\perp = U$.

How to find it

- $U \subseteq (U^\perp)^\perp$ is free; the danger is that $(U^\perp)^\perp$ is bigger.

- In finite dimension, count: $\dim(U^\perp)^\perp = n - (n - \dim U) = \dim U$.
- In general: split $v = u + w$ and show the $U^\perp$-part w is perpendicular to itself.

Proof. $U \subseteq (U^\perp)^\perp$: every $u \in U$ is perpendicular to every element of $U^\perp$. Conversely let $v \in (U^\perp)^\perp$, $v = u + w$ ($u \in U$, $w \in U^\perp$). Then $w = v - u \in (U^\perp)^\perp$ and $w \in U^\perp$, so $\langle w, w\rangle = 0$ and $w = 0$. □

Definition (Def. 48 / LADR 6.55)**.** For finite-dimensional U, the **orthogonal projection** is $P_U(u + w) = u$ ($u \in U$, $w \in U^\perp$). It is the unique projector with range $P = U$, null $P = U^\perp$.

- **Uniqueness:** if $P^2 = P$, range $P = U$, null $P = U^\perp$, then $u = Px$ gives $Pu = P^2x = u$, so $P(u + w) = u$.
- **Properties** (LADR 6.57), all from $v = P_Uv + (v - P_Uv)$:
  - P_U is linear; $P_U^2 = P_U$; range $P_U = U$; null $P_U = U^\perp$; $v - P_Uv \in U^\perp$.
  - $\|P_Uv\| \le \|v\|$ (Pythagoras).
  - $I - P_U = P_{U^\perp}$ if V is finite-dimensional ($V = U^\perp \oplus U$).
- **Any projector** ($P^2 = P$) has $V = \text{null } P \oplus \text{range } P$ (Prop. 91). Idea: Pv is the obvious range part, and $P(v - Pv) = Pv - P^2v = 0$. If $w = Pv \in \text{null } P$, then $w = P^2v = Pw = 0$.
- It is orthogonal only when null $P \perp$ range P: $P(x, y) = (0, y)$ is $P_{\text{span}(e_2)}$; $P(x, y) = (x + y, 0)$ is not orthogonal.

Proposition (Prop. 92 / LADR 6.57(i))
If $e_1, \dots, e_m$ is an orthonormal basis of U: $P_Uv = \sum_k \langle v, e_k\rangle e_k$.

How to find it

- The $U^\perp$-part of v is invisible to inner products with the e_k, so $\langle v, e_k\rangle = \langle P_Uv, e_k\rangle$: these are the coordinates of P_Uv.

Proof. $v = u + w$: $\langle v, e_k\rangle = \langle u, e_k\rangle$, so $P_Uv = u = \sum\langle u, e_k\rangle e_k = \sum\langle v, e_k\rangle e_k$ (Prop. 86 in U). □

Proposition (HW10 P5, LADR 6.61: best approximation)
For $u \in U$: $\|v - P_Uv\| \le \|v - u\|$, equality only at $u = P_Uv$. So the distance from v to U is $\|v - P_Uv\|$.

How to find it

- Split the error $v - u$ into a part $\perp U$ (no choice of u changes it) and a part in U (zero when $u = P_Uv$). Pythagoras adds them in squares.

Proof. $v - u = (v - P_Uv) + (P_Uv - u)$ with pieces in $U^\perp$ and U, so $\|v - u\|^2 = \|v - P_Uv\|^2 + \|P_Uv - u\|^2$. □

- **Bessel:** for any orthonormal $e_1, \dots, e_m$, $\sum |\langle v, e_k\rangle|^2 \le \|v\|^2$. The left side is $\|P_Uv\|^2$ with $U = \text{span}(e_k)$.
- **HW10 P7:** a projector is orthogonal iff $\|Pv\| \le \|v\|$ for all v. Idea for $\Leftarrow$: if $u \in \text{range } P$ and $w \in \text{null } P$ were not perpendicular, sliding u along w to $u + tw$ would shorten it while P still gives u, so P would stretch $u + tw$. (Details in the solutions.)
- **Computing P_Uv:** (1) Gram–Schmidt on U, then Prop. 92. (2) Or write $P_Uv = \sum c_iu_i$ for a basis u_i and solve $\langle v - \sum c_iu_i, u_j\rangle = 0$ for all j. Check that $v - P_Uv \perp$ each u_j.

Example (Ex. 71)
$P_U(x^3)$, $U = \text{span}(1, x)$, $\langle p, q\rangle = \int_0^1 pq$. With $e_1 = 1$, $e_2 = \sqrt{3}(2x - 1)$: $\langle x^3, e_1\rangle = \frac{1}{4}$, $\langle x^3, e_2\rangle = \frac{3\sqrt{3}}{20}$, so $P_U(x^3) = \frac{1}{4} + \frac{9}{20}(2x - 1) = \frac{9}{10}x - \frac{1}{5}$. Check: $x^3 - \frac{9}{10}x + \frac{1}{5}$ is perpendicular to 1 and x.

Traps

- Orthogonal vectors are independent only if nonzero.
- $P^2 = P$ does not make P orthogonal: $P(x, y) = (x + y, 0)$ has $\|P(1, 1)\| = 2 > \sqrt{2}$.
- $V = U \oplus U^\perp$ can fail for infinite-dimensional U: in ℓ^2, the finitely supported sequences have $U^\perp = \{0\}$. Also ℓ^2 has no orthonormal (Hamel) basis.
- Gram–Schmidt depends on the order and on the inner product; normalize only after subtracting.
- Riesz over $\mathbb{C}$: the coefficients are conjugated.

§2 Homework 8–10

Homework 8: generalized eigenvectors and Jordan form

HW8 P1. Let $D\colon \mathcal{P}_n(\mathbb{C}) \to \mathcal{P}_n(\mathbb{C})$ be differentiation.

(a) Prove that D is nilpotent and determine its nilpotency index.
(b) Determine the geometric and algebraic multiplicities of the eigenvalue 0.
(c) Find a basis with respect to which the matrix of D is a single Jordan block.

HW8 P2. Construct explicit matrices having the following spectral data.

(a) Eigenvalue 2 has algebraic multiplicity 3 and geometric multiplicity 1; eigenvalue -1 has algebraic multiplicity 2 and geometric multiplicity 2.
(b) The same algebraic multiplicities as in (a), but the geometric multiplicity of 2 is 2.

For each matrix, give its minimal polynomial.

HW8 P3. Let $J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

(a) Compute $\operatorname{null}(J - I)$, $\operatorname{null}((J-I)^2)$, and $\operatorname{null}((J-I)^3)$.
(b) Determine the generalized eigenspace $G(1, J)$ and the geometric and algebraic multiplicities of 1.
(c) Compute J^m for every positive integer m.

HW8 P4. Let N be nilpotent with Jordan form consisting of one block of size 3 and one block of size 2.

(a) Compute $\dim \operatorname{null} N^k$ for $k = 0, 1, 2, 3, 4$.
(b) Suppose instead that you are only told $\dim \operatorname{null} N = 2$, $\dim \operatorname{null} N^2 = 4$, $\dim \operatorname{null} N^3 = 5$. Explain how these numbers force the same two block sizes.

HW8 P5. (optional) Let $T \in \mathcal{L}(\mathbb{C}^6)$ have eigenvalues 1 and 2, and suppose

$$\dim \operatorname{null}(T - I) = 1, \quad \dim \operatorname{null}((T-I)^2) = 2, \quad \dim \operatorname{null}((T-I)^3) = 2,$$

while

$$\dim \operatorname{null}(T - 2I) = 2, \quad \dim \operatorname{null}((T-2I)^2) = 3, \quad \dim \operatorname{null}((T-2I)^3) = 4, \quad \dim \operatorname{null}((T-2I)^4) = 4.$$

Determine the Jordan normal form of T, up to reordering of the blocks.

HW8 P6. (optional) Suppose an operator has Jordan blocks $J_3(4)$, $J_2(4)$, $J_1(-1)$. Determine:

(a) its characteristic polynomial;
(b) its minimal polynomial;
(c) the geometric and algebraic multiplicities of each eigenvalue;
(d) whether it is diagonalizable.

HW8 P7. (optional) Prove that for a nilpotent operator N, $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ is the number of Jordan blocks of size at least k.

Homework 9: trace and inner products

HW9 P1. Let P be a projection on a finite-dimensional vector space, so that $P^2 = P$. Prove that $\operatorname{tr} P = \operatorname{rank} P$. You may choose a basis adapted to the decomposition $V = \operatorname{null} P \oplus \operatorname{range} P$.

HW9 P2. Let $T \in \mathcal{L}(\mathbb{C}^5)$ have minimal polynomial $m_T(x) = (x-2)^2(x+1)$.

(a) Determine all possible algebraic multiplicities of the eigenvalues 2 and -1.
(b) Determine all possible values of $\operatorname{tr} T$.
(c) For each possible trace, exhibit a Jordan normal form realizing it.

HW9 P3.

(a) Prove that every nilpotent operator on a finite-dimensional complex vector space has trace 0.
(b) Give a nonzero operator with trace 0 that is not nilpotent.
(c) Give two 2×2 matrices A, B for which $AB - BA \neq 0$, and verify directly that $\operatorname{tr}(AB - BA) = 0$.

HW9 P4. For each of the following formulas on $\mathbb{R}^2$, decide whether it defines an inner product. Justify your answer.

$$\langle x, y\rangle_1 = x_1y_1 + 2x_2y_2, \qquad \langle x, y\rangle_2 = x_1y_1 - x_2y_2, \qquad \langle x, y\rangle_3 = (x_1 + x_2)(y_1 + y_2).$$

HW9 P5. On $\mathcal{P}_2(\mathbb{R})$ define $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$.

(a) Prove that this is an inner product.
(b) Compute the norms of 1, x, and x^2.
(c) Determine which pairs among $1, x, x^2$ are orthogonal.

HW9 P6. Use the Cauchy–Schwarz inequality to prove that for all real numbers a, b, c, $(a+b+c)^2 \le 3(a^2+b^2+c^2)$. Determine exactly when equality holds.

HW9 P7. (optional) Prove the parallelogram identity $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$ in every real or complex inner-product space.

Homework 10: orthogonality, Gram–Schmidt, and projections

HW10 P1. On $\mathcal{P}_2(\mathbb{R})$ use the inner product $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$. Apply the Gram–Schmidt procedure to the ordered basis $1, x, x^2$ to obtain an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

HW10 P2. Let $W = \operatorname{span}\big((1,-1,1,0),(0,2,1,0)\big) \subseteq \mathbb{R}^4$ with the standard inner product. Find a basis of $W^{\perp}$.

HW10 P3. On $\mathcal{P}_2(\mathbb{R})$ use the same inner product as in Problem 1, and let $W = \operatorname{span}(x+1)$. Find a basis of $W^{\perp}$.

HW10 P4. Let U and W be subspaces of a finite-dimensional inner-product space V. Prove $(U \cap W)^{\perp} = U^{\perp} + W^{\perp}$.

HW10 P5. Let U be a subspace of a finite-dimensional inner-product space V, and let $e_1, \dots, e_m$ be an orthonormal basis of U.

(a) Prove that $P_U v = \sum_{j=1}^{m} \langle v, e_j\rangle e_j$.
(b) Prove the best-approximation property $\|v - P_U v\| \le \|v - u\|$ for every $u \in U$, and determine when equality holds.

HW10 P6. (optional) On $\mathcal{P}_1(\mathbb{R})$ use the inner product $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$. Find the unique polynomial $p \in \mathcal{P}_1(\mathbb{R})$ such that $\int_0^2 f(x)\,dx = \langle f, p\rangle$ for every $f \in \mathcal{P}_1(\mathbb{R})$.

HW10 P7. (optional) Let P be a projector on a finite-dimensional inner-product space, so $P^2 = P$. Prove that P is an orthogonal projection if and only if $\|Pv\| \le \|v\|$ for every v.

§3 Practice quizzes

Practice Quiz A (20 points)

1. Generalized eigenspaces. Let V be finite-dimensional, $n = \dim V$, and $T \in \mathcal{L}(V)$.

a) (3) Prove that $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$ for all k, and that if $\operatorname{null} T^m = \operatorname{null} T^{m+1}$, then $\operatorname{null} T^{m+j} = \operatorname{null} T^{m+j+1}$ for all $j \ge 0$.

b) (3) Prove that $\operatorname{null} T^n = \operatorname{null} T^{n+1}$, and deduce $G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$.

c) (2) Prove that $G(\lambda, T) \neq \{0\}$ if and only if λ is an eigenvalue of T.

d) (2) Let $A = \begin{pmatrix} 3 & 1 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$. Find $E(3, A)$, $G(3, A)$ and $G(2, A)$, and the Jordan form of A.

2. Trace.

a) (2) Let A be $m \times n$ and B be $n \times m$. Prove that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

b) (2) Deduce that similar matrices have the same trace, so that $\operatorname{tr} T$ for $T \in \mathcal{L}(V)$ does not depend on the basis.

c) (3) Let $V \neq \{0\}$ be a finite-dimensional real or complex vector space. Prove that there are no $S, T \in \mathcal{L}(V)$ with $ST - TS = I$. Then show that on $\mathcal{P}(\mathbb{R})$, with $D(p) = p'$ and $M(p) = xp$, $DM - MD = I$. Which hypothesis fails?

d) (3) Let $T \in \mathcal{L}(\mathbb{C}^4)$ have $m_T = (x-1)^2(x-3)$ and $\dim E(1, T) = 2$. Find the Jordan form of T and $\operatorname{tr} T$.

Practice Quiz B (20 points)

1. Counting Jordan blocks.

a) (2) Let $N = J_j(0)$ be a single nilpotent Jordan block of size j. Prove that $\dim \operatorname{null} N^k = \min(k, j)$ for $k \ge 0$.

b) (3) Let N be nilpotent on a finite-dimensional complex space, with Jordan form made of blocks $J_{j_1}(0), \dots, J_{j_r}(0)$. Prove that $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ is the number of blocks of size at least k, and deduce a formula for the number of blocks of size exactly k.

c) (3) Let $A = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$. Find the Jordan form of A and a basis of $\mathbb{C}^4$ in which A has that form.

d) (2) Let $J = J_3(2)$. Find a formula for J^k, $k \ge 1$.

2. Cauchy–Schwarz. Let V be a real or complex inner product space.

a) (4) Prove that $|\langle u, v\rangle| \le \|u\|\|v\|$, with equality if and only if u, v are linearly dependent.

b) (2) Deduce the triangle inequality $\|u + v\| \le \|u\| + \|v\|$.

c) (2) Show that the norm $\|(x, y)\|_\infty = \max(|x|, |y|)$ on $\mathbb{R}^2$ does not come from an inner product.

d) (2) Let $f\colon [0, 1] \to \mathbb{R}$ be continuous. Prove that $\left(\int_0^1 f\right)^2 \le \int_0^1 f^2$, with equality if and only if f is constant.

Practice Quiz C (20 points)

1. Orthonormal bases and Gram–Schmidt. Let V be an inner product space.

a) (2) Let $e_1, \dots, e_m$ be orthonormal. Prove $\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$ and deduce that $e_1, \dots, e_m$ is linearly independent.

b) (2) If $e_1, \dots, e_n$ is an orthonormal basis, prove $v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n$ for every v.

c) (3) Let $v_1, \dots, v_n$ be linearly independent and let $e_1, \dots, e_{j-1}$ be orthonormal with $\operatorname{span}(e_1, \dots, e_{j-1}) = \operatorname{span}(v_1, \dots, v_{j-1})$. Let $w_j = v_j - \sum_{k<j} \langle v_j, e_k\rangle e_k$. Prove that $w_j \neq 0$, that $e_j = w_j / \|w_j\|$ is orthogonal to $e_1, \dots, e_{j-1}$, and that $\operatorname{span}(e_1, \dots, e_j) = \operatorname{span}(v_1, \dots, v_j)$.

d) (3) On $\mathcal{P}_2(\mathbb{R})$ let $\langle p, q\rangle = p(0)q(0) + p(1)q(1) + p(2)q(2)$. Show that this is an inner product, and apply Gram–Schmidt to $1, x, x^2$.

2. Orthogonal complements and projections. Let V be a finite-dimensional inner product space and U a subspace.

a) (3) Prove that $V = U \oplus U^{\perp}$, and deduce $\dim U^{\perp} = \dim V - \dim U$.

b) (2) Prove that $(U^{\perp})^{\perp} = U$.

c) (2) Prove that $\|P_U v\| \le \|v\|$ for every $v \in V$.

d) (3) In $\mathbb{R}^3$ with the dot product let $U = \operatorname{span}\big((1,1,0), (1,0,1)\big)$. Find an orthonormal basis of U, $P_U(1,2,3)$, a basis of $U^{\perp}$, and the distance from $(1,2,3)$ to U.